# Technical Document #1002: Computer Vision - Comprehensive Overview

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Object detection identifies and localizes objects in images. YOLO and Faster R-CNN are popular real-time object detection architectures.

Pose estimation detects human body keypoints in images. It is used in action recognition, animation, and sports analysis.

Visual question answering (VQA) answers questions about images. It requires both visual understanding and language comprehension.

Image super-resolution reconstructs high-resolution images from low-resolution inputs. GAN-based methods produce photorealistic upscaled images.

Optical character recognition (OCR) converts images of text into machine-readable text. It is used in document digitization and automation.

Face recognition identifies or verifies faces in images. DeepFace and FaceNet achieve near-human accuracy on face verification tasks.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

### Key Takeaways:

- Image generation creates new images from random noise or text descriptions.
- Video understanding analyzes temporal dynamics in video sequences.
- Visual question answering (VQA) answers questions about images.